

Power & Propulsion Systems for UVs — 1D IC-Engine Simulation & P1 Hybrid Energy Management

Exam project for **Power and Propulsion Systems**, Master's in Autonomous Vehicle Engineering, Università degli Studi di Napoli Federico II (UniNa).

Overview

Two connected studies on vehicle powertrains:

- **Exercise 1 — 1D IC-Engine simulation (GT-Power / GT-SUITE):** a four-cylinder, port-fuel-injected, spark-ignition engine model with predictive SI combustion, knock model, fuel-film dynamics and a wall-temperature solver. Baseline geometry scaled by assigned multipliers (bore $\times 0.97$, stroke $\times 1.04$). Four parametric studies sweep engine speed, air–fuel ratio (rich/lean), spark advance, and throttle/load, interpreted through mean-effective-pressure theory (BMEP, IMEP, BSFC, torque, power, volumetric efficiency).
- **Exercise 2 — P1 hybrid energy management (MATLAB/Simulink, Powertrain Blockset):** a P1 hybrid-electric vehicle with an Equivalent Consumption Minimization Strategy (ECMS) controller. Fuel economy, battery SOC and tailpipe emissions are evaluated against Euro 6e limits and the EU 2025 CO₂ target across WLTC, Artemis Rural and Artemis Motorway cycles, plus a $\pm 25\%$ vehicle-mass sensitivity study.

This repository contains the **GT-Power engine model** (Exercise 1) together with the full report and presentation covering **both** exercises. The Simulink P1 hybrid model is documented in the report/slides.

Requirements

- **GT-SUITE / GT-Power** to open and run the `.gtm` model files and view `.gdx` results (developed with the course template `SI_4cyl_SITurb_PortFuelFilm_TWallSoln`).
- A PDF viewer for the report, guide and task; PowerPoint (or compatible) for the presentation.
- *(Exercise 2 only, not included here)* MATLAB/Simulink with Simscape and Powertrain Blockset.

Repository contents

File / folder	Description
Report.pdf	Full technical report covering Exercise 1 (ICE) and Exercise 2 (P1 hybrid).
Presentation.pptx	Project presentation slides.
Task.pdf	Assignment brief / exercise instructions.
Guide.pdf	GT-Power modelling guide / reference.
Engine Model/	GT-Power model and results files (see below).

Engine Model/

File	Description
Model.gtm, Engine_Model.gdx	Baseline engine model and results.
SI_4cyl_SITurb_PortFuelFilm_TWall Soln_spark.gtm	Course template engine model.
CASE 1/Harshal_Engine_Model_Case1.gtm (+ .gdx)	Engine-speed sweep (1000–5000 rpm).
CASE 2/..._Case2_rich.gtm, ..._Case2_lean.gtm (+ .gdx)	Air–fuel ratio sweep (rich AFR 11.76 / lean AFR 17.64).
CASE 3/Harshal_Engine_Model_Case3.gtm (+ .gdx)	Spark-advance sweep at 3000 rpm.
CASE 4/Harshal_Engine_Model_Case4.gtm (+ .gdx)	Throttle-position / load sweep.

.gtm = GT-SUITE model file · .gdx = GT-SUITE results/data file.

How to use

1. **Read the report** — start with [Report.pdf](#) for the full methodology and results; [Presentation.pptx](#) is the summary.
2. **Open a model** — launch GT-SUITE/GT-Power and open the [.gtm](#) for the case you want (start from [Engine Model/Model.gtm](#), then the [CASE 1–4](#) folders).
3. **Run / inspect results** — run the model in GT-Power, or open the matching [.gdx](#) to view the pre-computed results used in the report.
4. **Reference material** — [Task.pdf](#) is the assignment; [Guide.pdf](#) supports the GT-Power setup.

Key results

- **Exercise 1:** torque/BMEP peak near 3500–4000 rpm while power keeps rising to 5000 rpm; minimum BSFC (~240 g/kWh) near 3000 rpm; the rich-power vs. lean-economy trade-off reproduced across the AFR sweep.
- **Exercise 2:** ~5.5 L/100km on WLTC Class 3, charge-sustaining SOC balance via a tuned ECMS equivalence factor ($s = 3.23$); HC/CO/NO_x well within Euro 6e, but mild P1 hybridization alone is insufficient to meet the EU 2025 CO₂ target.

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